

Revised Generation Interconnection System Impact Study Report

For Queue Project Q25-2497

Baytown Energy Center 100 kV
January 01, 2026

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q25-2497. The Interconnection Customer proposes a 30.0 MW battery energy storage facility interconnecting at the Baytown Energy Center 100 kV substation in the Illinois area. The request was received on January 18, 2025. The study evaluated system intact and single contingency conditions using the regional planning case, including 436 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$62,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: ZONA LIBRE 6007 / GEORGE NEAL SOUTH 230 kV
Contingency: GALION GENERATING STA / TREGO 500 kV
Loading: 104.8% (Rating: 2,483 MVA)
DFAX: 6.3%
Pre-project loading: 104.7%
2. Facility: MYRTLE POINT / GEORGE NEAL S 230 kV
Contingency: TREGO / GAION GENERATING STA SUBSTATION 500 kV Line
Loading: 102.7% (Rating: 2,517 MVA)
DFAX: 6.3%
Pre-project loading: 102.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ZONA LIBRE 6007 / GEORGE NEAL SOUTH 230 kV	Reconductor line	\$9,607,000	0% (\$36,000)
MYRTLE POINT / GEORGE NEAL S 230 kV	Reconductor line	\$6,868,000	0% (\$26,000)
	Total allocated to project		\$62,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.